Demand Estimation in the Cola Market:

An Aggregate Logit Analysis of Price Elasticities and Promotion Uplifts

1. Introduction

The carbonated soft drink market features intense competition between Coke and Pepsi through price and promotion strategies. Building on SWP 2's reduced-form analysis, this paper estimates an aggregate logit demand model to analyze consumer choice among cola products, with a specific focus on how product dimensions influence demand patterns. The analysis strategically examines: (1) caffeinated vs. caffeine-free formulations (focusing on dominant caffeinated variants), (2) brand-level vs. package-level aggregation, and (3) regular vs. diet formulations within the cola category. This approach allows estimation of how consumers trade off across product attributes while maintaining model tractability.

The primary objectives are to: (1) estimate price elasticities correcting for endogeneity across product dimensions, (2) quantify promotional effectiveness at different aggregation levels, (3) assess substitution patterns across brand and package formats, and (4) compare structural estimates with SWP 2's reduced-form results.

I implement Berry's (1994) inversion to transform market shares into a linear estimating equation, addressing price endogeneity through instrumental variables. The analysis employs format-specific IV strategies: Hausman instruments for brand-level analysis and aluminum price instruments for 12-can analysis, recognizing that different product tiers require different identification approaches.

Section 2 shows the utility function that includes all main drivers of utility, Section 3 details the data and product dimension definitions. Section 4 presents the econometric strategy with format-specific IV approaches. Section 5 reports elasticities and promotional uplifts. Section 6 as conclusion compares results with SWP 2, and concludes with implications for product portfolio management.

2. The Aggregate Logit Demand System

For consumer *i* choosing product *j* in market *t* (store-week), indirect utility is:

$$u_{ijt} = \delta_{jt} + \epsilon_{ijt} = \alpha p_{jt} + \sum_b \beta_b D_j^b + \gamma' X_{jt} + \xi_{jt} + \epsilon_{ijt}$$

Where:

- p_{jt}: Price per liter (endogenous)
- D_{jb}: Brand dummies (Coke Classic, Diet Coke, Pepsi, Diet Pepsi)
- X_{jt}:Promotional indicators(display.major, display.minor, feature.large, feature.medium, coupon)
- ξ_{jt}: Unobserved quality (demand shock)
- ϵ_{ijt}: Type-I Extreme Value error

The same utility specification is applied to a restricted subset of the market, focusing exclusively on the best-selling 12-can format while processing Package level analysis, The outside good is redefined to include all non-12-can beverage alternatives. Aluminum price is used as the primary instrumental variable (rather than spatial instruments), as it directly affects can production costs but is uncorrelated with local demand shocks for this specific format. Following Berry (1994), market shares are inverted to obtain the linear estimating equation. Own- and cross-price elasticity formulas are also provided in Appendix A.

3. Data and Market Definition

3.1 Data Sources (HU-Box Files)

The analysis draws on IRI scanner data retrieved from the HU-Box repository. The primary dataset (cola.coke.pepsi.csv) contains a product-store-week panel spanning 2004–2006. Supplementary files include store characteristics, weekly trip counts, total revenue records, and instrumental variables such as aluminum prices, sugar prices, and local weather data.

3.2 Level Demand

Market is defined as a unique store (IRI_KEY) × week (WEEK) combination across two DMAs (Eau Claire and Pittsfield), 2004 - 2006. Which is invalid of 1964 out of 1965 stores. Initial merge produced 78,052 observations. We removed 35 (0.05 % of sample) observations where cola-purchasing trips exceeded total store trips, an impossibility indicating scanner recording errors. The final estimation sample contains 75,317 observations.

3.3 Product Dimension Strategy and Selection (Inside Goods)

The cola market exhibits differentiation across multiple dimensions: caffeine content, dietary formulation, package size, and format. To balance empirical tractability with economic relevance, I implement a strategic dimension reduction approach:

3.3.1 Caffeine Content Focus

Preliminary analysis reveals caffeine-free variants constitute less than 3% of total cola sales and exhibit systematically different demand patterns (brand fixed effects approximately 30% more negative). Following the principle of focusing on economically significant market segments, the analysis concentrates on caffeinated products, which drive category volume and represent the core Coke-Pepsi competition.

3.3.3 Dietary Formulation Consideration

Regular and diet variants are included as distinct products to capture baseline preference differences. However, the aggregate logit's IIA property imposes symmetric substitution patterns—the cross-elasticity matrix shows Coke Classic draws equal share from Diet Coke as from Pepsi (−0.04). Thus, while diet status affects baseline utility, it does not generate attribute-based substitution in this specification.

3.3.4 Package Format Selection

The 12-can format is selected for disaggregated analysis because: (1) aluminum price instruments directly affect can production costs, providing strong IV identification; (2) multipacks represent strategic stock-up purchases; (3) format homogeneity simplifies preference estimation.

4. Econometric Strategy & Endogeneity Correction

4.1 Endogeneity Problem

Naive OLS estimates (ignoring price endogeneity yield a semi-elasticity of -1.44 (SE: 0.012) in the parsimonious specification, implying an own-price elasticity of -1.58 at mean prices. This aligns closely with the SWP 2 log-log estimate of -1.25, validating that the brand-level aggregation preserves the demand structure identified in prior analysis. Further OLS Estimation Results see Appendix C: OLS Benchmarks.

However, comparing specifications reveals multicollinearity concerns: adding feature advertisements attenuates the price coefficient toward zero (-1.23), suggesting promotions correlate with unobserved demand shocks. The R-squared of 0.97 reflects the explanatory power of store fixed effects, but the residual bias indicates OLS is inconsistent for price elasticity estimation.

Table 1: OLS Estimation

Specification	Price Coefficient	Implied Elasticity*	R-squared
OLS Full	-1.229	-1.35	0.970
OLS without features	-1.440	-1.58	0.968

While these OLS estimates provide a useful benchmark, they may suffer from endogeneity bias. Price endogeneity arises from three sources: (1) Supply-side simultaneity: retailers set prices observing demand shocks ξ_{jt} unobservable; (2) Promotional coordination: temporary price cuts coincide with unobserved shelf-space allocations; (3) Quality omitted variable bias: premium positioning correlates with both price and unobserved brand equity.

To address endogeneity of p_{jt} (correlation with unobserved quality ξ_{jt}), I employ three instrumental variable strategies.

4.2 Instrumental Variables Strategy

For brand-level analysis, Hausman instruments (prices in other market with same week, brand and size) leverage spatial price correlations. For 12-can multipack analysis, aluminum spot prices provide a cost-based instrument directly tied to can production. This format-specific strategy is necessary because Hausman instruments yield implausibly high elasticities ($\eta = -4.75$) at the SKU level, likely due to weak instrument bias.

First-Stage Results: Table 2 presents first-stage regression results. Instruments show strong predictive power: price in other markets has a coefficient of 0.910 (t=574.5), indicating nearly one-to-one spatial price transmission. Aluminum price is statistically significant but economically minimal. The joint F-statistic of 111,692 ($p<0.001$) decisively rejects the weak instruments hypothesis. Promotional variables show negative coefficients (e.g., large feature: -0.172), confirming that promotions correlate with temporary price reductions, creating endogeneity bias. First-Stage Instrument Joint Significance Test Results see Appendix D.

Table 2: First-Stage Regression Results

Dependent variable: Price per Liter

Variable	Coefficient	(Std. Error)	t-value
Instrumental Variables			
Aluminum Price	-0.0000185***	(0.0000022)	-8.58
Price in Other Markets	0.9096***	(0.0016)	574.49
Avg. Price Other Stores	0.4495***	(0.0156)	28.76

Variable	Coefficient	(Std. Error)	t-value
Promotional Variables			
Major Display	-0.0629***	(0.0018)	-35.50
Minor Display	-0.0657***	(0.0026)	-24.81
Large Feature	-0.1723***	(0.0029)	-60.23
Medium Feature	-0.1643***	(0.0028)	-58.24
Coupon	-0.0446***	(0.0077)	-5.82
Brand Dummies			
Diet Pepsi	0.0200***	(0.0021)	9.35
Pepsi	0.0162***	(0.0021)	7.77
Diet Coke	0.0000276	(0.0019)	0.01
Model Statistics			
Observations	47,442		
R^2	0.914		
F-statistic	21,790***		

4.3 IV results: Specification Comparison and Preferred Model

Brand-Level Analysis (Hausman IV): The preferred specification yields a price coefficient of -1.267 (SE=0.013), translating to a mean own-price elasticity of -1.40. This is slightly larger than the OLS estimate (-1.35), suggesting OLS modestly understates price sensitivity. Promotional effects remain positive and significant but are generally smaller than OLS estimates. All brand fixed effects show negative coefficients as expected. See Appendix E.

12-Can Package Analysis (Aluminum IV): The aluminum IV specification yields a price coefficient of 3.18, translating to a mean own-price elasticity of -2.71, which approximately double the brand-level magnitude and substantially more plausible than the Hausman IV result for SKUs (-4.75). This confirms that multipacks face more elastic demand, consistent with stockpiling behavior.

Diagnostic Tests:

Formal diagnostic tests confirm the necessity and validity of both IV approaches. For brand-level analysis, the Wu-Hausman test rejects the null hypothesis of exogeneity (χ^2=119.1, p<0.001), indicating OLS inconsistency. The Cragg-Donald F-statistic of 327,995 far exceeds the Stock-Yogo critical value of 16.38, confirming strong instruments. The exclusion restriction for Hausman instruments is theoretically justified: prices in geographically separate markets affect local sales only through their impact on local prices. While common supply factors create spatial price correlations, demand shocks in other markets are unlikely to directly influence local consumer preferences.

For aluminum instruments, the exclusion restriction relies on aluminum prices affecting local demand only through production costs. Global aluminum markets are sufficiently disconnected from local cola preferences to satisfy this condition, particularly given aluminum's minimal share in final consumer budgets. Alternative instrument sets produce less credible results: cost shifters for brand-level analysis yield implausible elasticities (-5.100), while spatial instruments for 12-can analysis produce unrealistically high estimates (-4.75). This pattern confirms the superiority of the tiered IV approach that matches instrument type to product aggregation level.

Specification Comparison: Alternative IV specifications yield less credible results. Cost shifters for brand-level analysis produce implausible elasticities (-5.100), while spatial instruments for 12-can analysis yield unrealistically high estimates (-4.75). This pattern confirms the superiority of matching instrument type to product aggregation level.

5. Elasticities & Promotion Uplifts

5.1 Brand level

5.1.1 Own-Price Elasticities

Table 3 presents own-price elasticities estimated using the IV-Hausman specification (instrumenting with cross-market prices) on the brand-aggregated sample of 47,442 observations

Table 3: Own-Price Elasticities by Brand (IV-Hausman)

Brand	Mean Price ($/L)	Mean Market Share	Own-Price Elasticity	Observations
Coke Classic	1.13	0.505%	-1.42	12,527
Diet Coke	1.12	0.458%	-1.42	12,327
Pepsi	1.08	0.473%	-1.37	11,218
Diet Pepsi	1.07	0.355%	-1.36	11,370

The IV-Hausman estimation yields a mean own-price elasticity of -1.40, indicating elastic demand where a 10% price increase reduces choice probability by approximately 14%. The estimates are remarkably consistent across brands, ranging narrowly from -1.36 (Diet Pepsi) to -1.42 (Coke Classic and Diet Coke). This similarity reflects both the IIA property of the logit specification and comparable price sensitivity across the four major cola variants.

Notably, despite commanding a 5 - 6% price premium ($1.13 vs. $1.07 - $1.08), Coke Classic exhibits the same elasticity magnitude as its lower-priced competitors, suggesting that brand equity does not substantially alter price sensitivity at the aggregate level. Market shares align with this hierarchy: Coke Classic holds the largest individual share (0.505%), followed by Pepsi (0.473%) and Diet Coke (0.458%), with Diet Pepsi trailing at 0.355%.

5.1.2 Cross-Price Elasticities

Cross-price elasticities are small (range: 0.005 - 0.007), indicating limited substitution between brands. For example, a 10% price increase by Pepsi increases Coke Classic demand by only 0.07%. The IIA restriction manifests in the identical cross-elasticities across competitor pairs: the model predicts that consumers are equally likely to switch from Pepsi to Coke Classic as to Diet Coke following a price change, despite the likely closer substitutability within brand families (Coke Classic compared with Diet Coke vs. Coke Classic compared Pepsi). This mechanical symmetry represents a key limitation of the aggregate logit specification, motivating nested structures for future research.

Table 4: Cross-Price Elasticities

Brand	Coke Classic	Diet Coke	Pepsi	Diet Pepsi
Coke Classic	-1.42	-0.007	-0.007	-0.005
Diet Coke	-0.007	-1.42	-0.007	-0.005
Pepsi	-0.007	-0.007	-1.37	-0.005
Diet Pepsi	-0.007	-0.007	-0.007	-1.36

5.1.3 Promotional Uplifts

The IV-Hausman specification reveals substantial variation in promotional effectiveness across instruments. Table 5 reports semi-elasticities (log-odds coefficients), odds ratios, and implied share-point lifts at mean market share (0.45%).

Table 5: Promotional Effects (IV-Hausman Estimates)

Promotion Type	Coefficient	Odds Ratio	% Change in Odds	Share Point Lift
Major Display	0.570***	1.768	+76.8%	0.00347
Minor Display	0.632***	1.881	+88.1%	0.00398
Large Feature	1.244***	3.469	+246.9%	0.01115
Medium Feature	0.817***	2.264	+126.4%	0.00571
Coupon	1.858***	6.413	+541.3%	0.02444

The coefficients represent semi-elasticities in the log-odds space: a Display Major increases the odds of choosing the brand by a factor of 1.77 (76.8% increase), while the presence of a coupon roughly sextuples the odds (odds ratio 6.41, implying a 541% increase in relative odds compared to the outside good).

Translating to absolute market share impacts at the sample mean (0.45% brand share), the promotional effects range from 0.35 percentage points (Display Major) to 2.44 percentage points (Coupon). This implies that a coupon promotion increases a brand's share of total store traffic from the baseline 0.45% to approximately 2.89%, a more than six-fold proportional increase within the cola category.

5.2 Package level- Best Seller

The 12-can multipack represents the primary battleground for household stock-up trips. For this analysis, I employ aluminum price as cost shifter instruments rather than Hausman price instruments, as aluminum costs directly affect can production but are orthogonal to local demand shocks. The IV estimation yields a price coefficient of 3.18, translating to a mean own-price elasticity of -2.71.

5.2.1 Own-Price Elasticity and Demand Intensity

The aluminum IV specification produces economically plausible elasticities that are approximately twice the brand-level magnitude (-2.71 vs. -1.40), confirming that multipack purchases exhibit greater price sensitivity.

Table 6: Own-Price Elasticity (cost shifter)

Brand	Mean Price ($/L)	Mean Market Share	Own-Price Elasticity	Observations
Coke Classic	0.836	1.46%	2.63	1,964
Diet Coke	0.836	1.34%	2.63	1,960
Diet Pepsi	0.866	0.792%	2.74	1,943
Pepsi	0.873	0.998%	2.76	1,962

This aligns with the stock-up nature of multipack consumption, where consumers actively compare value across larger inventory investments.

5.2.2 Cross-Price Substitution Patterns

Cross-price elasticities range from -0.02 to -0.04 (Table 7), representing a fivefold increase over brand-level estimates but remaining economically moderate. This suggests that while 12-can consumers are more price-sensitive than brand-average shoppers, they still exhibit meaningful brand loyalty within the multipack format.

Table 7: Cross-Price Elasticity Matrix (12-Can)

Brand	Coke Classic	Diet Coke	Pepsi	Diet Pepsi
Coke Classic	0.00	-0.04	-0.03	-0.02
Diet Coke	-0.04	0.00	-0.03	-0.02
Pepsi	-0.04	-0.04	0.00	-0.02
Diet Pepsi	-0.04	-0.04	-0.03	0.00

5.2.3 Promotional Uplifts for 12- can package

Table 8 presents promotional semi-elasticities for the 12-can specification, revealing substantially stronger treatment effects compared to brand-level aggregates. The IV estimates using the appropriate aluminum cost instrument indicate that promotional instruments exert significantly greater influence on SKU-level choice probabilities than on brand-level aggregation, with coefficients ranging from 1.103 to 1.668. These effects are roughly 1.5 to 2 times larger than those observed in the brand-level Hausman IV analysis (0.570 to 1.858).

Table 8: Promotional Uplifts for 12- can package

Promotion Type	Coefficient	Odds Ratio	% Change in Odds	Share Point Lift
Major Display	1.103	3.01	+201%	0.0231
Minor Display	1.357	3.88	+288%	0.0331
Large Feature	1.629	5.10	+410%	0.0471
Medium Feature	1.279	3.59	+259%	0.0298
Coupon	1.668	5.30	+430%	0.0494

6. Conclusion

This paper estimates an aggregate logit demand model for cola products, employing format-specific instrumental variable strategies to address price endogeneity. The analysis focuses on caffeinated products (comprising 97% of category sales) and examines demand at both brand and 12-can package levels.

Contrasting these findings with SWP 2 reveals both methodological convergence and significant advances. The log-log model in SWP 2 estimated a pooled price elasticity of -1.24, which aligns closely with our OLS (-1.23) and IV (-1.27 to -1.40) brand-level estimates. This proximity suggests that store fixed effects in SWP 2 partially mitigated endogeneity bias. However, the fundamental distinction lies in the modeling approach: SWP 2 estimated sales volume conditional on category purchase, while the structural logit model estimates choice probabilities relative to an outside good. This enables the crucial estimation of cross-price elasticities (0.005–0.007 at brand level; 0.02–0.04 at SKU level), a dimension SWP2’s reduced-form framework could not capture. While SWP2 successfully identified highly elastic demand for 12-can multipacks (ranging from -2.60 to -3.02), it could not quantify how sales shift between competitors. The logit model reveals that substitutability between Coke and Pepsi is statistically significant but economically modest, with cross-elasticities an order of magnitude smaller than own-price effects.

Key empirical findings include: Brand-level elasticity of -1.40 (Hausman IV), indicating moderately elastic demand; Package-level elasticity of -2.71 (Aluminum IV) for 12-can multipacks, confirming approximately twice the price sensitivity compared to brand aggregates which consistent with SWP2's findings for this format; Promotional effects are 2 - 3 times stronger at the SKU level than at the brand level, particularly for coupons and feature advertisements; Limited brand switching, with cross-price elasticities under 0.04, suggesting strong brand loyalty within the cola category.

The structural framework provides several advantages over the reduced-form approach. First, it explicitly accounts for the outside good (88.21% market share), grounding choice in a complete market context. Second, the Berry inversion yields exceptionally strong identification (F-statistic = 111,692), addressing endogeneity more robustly than fixed effects alone. Third, the model generates a full elasticity matrix, enabling simulation of market share redistribution—a capability essential for merger analysis, optimal pricing, and promotional planning.

Managerial Implications support a tiered strategy informed by consumption occasion:

Multipack Pricing: With elasticities approximately twice the brand average (-2.71), multipacks operate in a fiercely competitive, price-sensitive segment. Aggressive pricing in this "stock-up" occasion risks significant profit erosion.

Promotion Allocation: The dramatically stronger promotional coefficients at the SKU level justify concentrating trade spending on specific high-volume formats, particularly coupons and large features for multipacks, and in-store displays for single-serve impulse buys.

Portfolio Coordination: The IIA property generates a counterintuitive implication: promotional gains for one product (e.g., Coke Classic) draw share equally from all rivals, including its own brand variants. While this is likely an artifact of the model's restriction and overstates cross-brand substitution, it underscores the need to manage promotions holistically to mitigate unintended cannibalization within a brand portfolio.

In conclusion, this analysis demonstrates that a structural demand model, even with simplifying assumptions like IIA, shifts the analytical focus from explaining historical sales to enabling forward-looking scenario planning. While the aggregate logit confirms the intense price competition identified in SWP2, its core value lies in quantifying substitution patterns and providing the necessary framework for counterfactual policy simulation—capabilities fundamental to strategic decision-making in a concentrated, competitive market.

Appendix

Appendix A: Theoretical Framework

A.1 Logit market share equation
The logit choice probability equals the market share:

$$s_{jt} = \frac{\exp(\delta_{jt})}{\sum_{k=0}^{J} \exp(\delta_{kt})} = \frac{\exp(\delta_{jt})}{1 + \sum_{k=1}^{J} \exp(\delta_{kt})}, \qquad j = 1, \ldots, J,$$

and the outside good share is:

$$s_{0t} = \frac{\exp(\delta_{0t})}{\sum_{k=0}^{J} \exp(\delta_{kt})} = \frac{1}{1 + \sum_{k=1}^{J} \exp(\delta_{kt})}$$

A.2 Berry Inversion & Estimating Equation
The choice probability yields market share:

$$s_{jt} = \frac{\exp(\delta_{jt})}{1+\sum_k \exp(\delta_{kt})}$$

Taking log-odds relative to the outside good (Berry inversion) yields:

$$\ln(s_{jt}) - \ln(s_{0t}) = \delta_{jt}$$

Substituting the linear index for mean utility gives the aggregate logit estimating equation:

$$\ln(s_{jt}) - \ln(s_{0t}) = \alpha p_{jt} + \sum_b \beta_b D_{jb} + \gamma' X_{jt} + \xi_{jt}$$

The dependent variable ln(*sjt*/*s*0*t*) represents the log-odds of choosing product *j* over the outside good (non-cola alternatives). This differs fundamentally from SWP 2's ln(*Qjt*) , which is conditioned on category participation.
A.3 Elasticity Formulas
Unlike SWP 2's constant elasticity, logit elasticities vary by market share:

Own-price:

$$\varepsilon_{jkt} = \alpha \cdot p_{kt} \cdot s_{kt}$$

Cross-price:

$$\varepsilon_{jjt} = \alpha \cdot p_{jt} \cdot (1 - s_{jt})$$

These elasticities exhibit the Independence of Irrelevant Alternatives (IIA) cross-elasticities are proportional to the competitor's share, not the similarity of products. This restriction will be evaluated in Section 5.

Appendix B: Brand Distribution

Brand	COKE CLASSIC	DIET COKE	DIET PEPSI	PEPSI
Observation	12922	12789	11963	11644

Appendix C: OLS Benchmark

Determinants of Product Market Share with features

Variable	Coefficient	Standard Error	t-value
Marketing Variables			
Major Display	0.610***	0.013	46.12
Minor Display	0.703***	0.020	35.36
Large Feature	1.274***	0.022	58.59
Medium Feature	0.839***	0.022	38.94
Coupon	1.870***	0.057	32.90
Price Sensitivity			
Price per Liter	-1.229***	0.012	-100.68
Brand Fixed Effects			
Coke Classic	-5.128***	0.029	-177.76
Diet Coke	-5.215***	0.029	-181.01
Pepsi	-5.237***	0.029	-183.69
Diet Pepsi	-5.488***	0.028	-193.26
Residual standard error: 1.163 on 49296 degrees of freedom			
Multiple R-squared: 0.9701, Adjusted R-squared: 0.9701			
F-statistic: 7.269e+04 on 22 and 49296 DF, p-value: < 2.2e-16			

Determinants of Product Market Share without features

Variable	Coefficient	Standard Error	t-value
Price Variable			
Price per Liter	-1.440***	(0.012)	-117.65
Marketing Variables			
Major Display	0.783***	(0.013)	58.02
Minor Display	0.880***	(0.020)	42.93

Coupon	1.603***	(0.059)	27.13
Brand Fixed Effects			
Coke Classic	-4.723***	(0.029)	-161.36
Diet Coke	-4.813***	(0.029)	-164.59
Pepsi	-4.839***	(0.029)	-167.16
Diet Pepsi	-5.102***	(0.029)	-176.75
Store Fixed Effects	Included		
Model Statistics			
Observations	49,318		
R-squared	0.968		
Adjusted R-squared	0.968		
Residual Std. Error	1.212 (df = 49,298)		
F-statistic	73,440 (df = 20; 49,298)		

Appendix D: First-Stage Instrument Joint Significance Test

Test Statistic	Value	Degrees of Freedom	p-value	Critical Value (10% max size)
Partial F-statistic	111,692	(3, 47,418)	<0.001	16.38
Restricted RSS	8,926.8	df = 47,421		
Unrestricted RSS	1,106.7	df = 47,418		
Sum of Squares Reduction	7,820.2	Δdf = 3		

Appendix E: IV estimates using different instruments for brand level

Variable	(1) Cost Shifters	(2) Other Markets	(3) Leave-One-Out	(4) Full Model
Price Elasticity				
Price per Liter	-5.100*** (0.496)	-1.267* (0.013)**	-3.167*** (0.133)	-1.276*** (0.013)
Brand Fixed Effects				
Coke Classic	0.387 (0.708)	-5.060*** (0.030)	-2.368*** (0.192)	-5.047*** (0.030)
Diet Coke	0.273 (0.705)	-5.156*** (0.030)	-2.468*** (0.191)	-5.143*** (0.030)

Diet Pepsi	-0.201	-5.382***	-2.842***	-5.370***
	(0.679)	(0.029)	(0.184)	(0.029)
Pepsi	0.097 (	-5.154***	-2.567***	-5.141***
	0.685)	(0.029)	(0.186)	(0.029)
Display Activities				
Major Display	-0.874*	0.570***	-0.133*	0.566***
	(0.192)**	(0.014)	(0.053)	(0.014)
Minor Display	-0.845*	0.632***	-0.072	0.628***
	(0.201)**	(0.020)	(0.058)	(0.020)
Feature Advertising				
Large Feature	-0.178	1.244* **	0.547***	1.240***
	(0.190)	(0.022)	(0.056)	(0.022)
Medium Feature	-0.598***	0.817***	0.120***	0.814***
	(0.188)	(0.022)	(0.056)	(0.022)
Promotion				
Coupon	1.132***	1.858***	1.501***	1.857***
	(0.137)	(0.058)	(0.074)	(0.058)
Model Statistics				
R^2	0.909	0.970	0.955	0.970
Weak IV F-stat	45.39	327,995	632.9	111,692
Observations	49,318	47,442	49,318	47,442

Appendix F: IV Diagnose

Test	Model 1	Model 2	Model 3	Model 4
Weak Instruments	F=45.39***	F=327,995*	F=632.9***	F=111,692***
Wu-Hausman	χ^2=185.8***	χ^2=119.1***	χ^2=325.5***	χ^2=168.9***
Sargan	χ^2=56.66***	NA	NA	χ^2=248.5***
Precision	Low (large SEs)	Very high	Moderate	Very high